FROM THE

EDITOR-IN-CHIEF

The moon belongs to everyone, but the brilliance of Indian space scientists has given us a special sense of ownership. My mind too goes back to a personal memory. The moon was the subject of a high school project I had laboured over with much love. Neil Armstrong's celebrated touchdown was still years away, but the Soviet Union's pioneering Luna-2 mission was about to make history, and space exploration had fired everyone's imagination. Back in my school, I had memorised the names of the lunar maria—the 'seas' of the moon. Most of them turn out to be located on its visible side. And funnily, it is on the dark side, permanently turned away from us, that Chandrayaan-1's Moon Impact Probe discovered water for the first time in 2008. It is also there, in the vicinity of the lunar South Pole, that India finally set down its footprints with Chandrayaan-3 on August 23. As fascinating new data flows back to Earth, we honour our scientists with an in-depth journalistic exploration of our own. Our cover story this week tells the inside story of how the Indian Space Research Organisation (ISRO) overcame the disaster of Chandrayaan-2 in 2019 to pull off a feat that will be entered into humanity's space logbook as historic.



LUNAR GLOW Chengappa with Veeramuthuvel and Kalpana

Despite literally dozens of moon missions since Luna-2 reached the moon's surface in 1959, becoming the first human-made object to make contact with another celestial body, ISRO's feat is remarkable. It accomplished a soft landing, a delicate task at which even Japan and Israel failed recently. And even if China's Chang'e 4 was the first to land on the far side, Chandrayaan-3 was the first conqueror of the actual South Pole region. It's a mysterious zone. The craters littered around this area, whose depths have not seen sunlight for over two billion years, make it difficult terrain. Also, it's subject to high temperature fluctuations unlike the moon's Equator, which receives constant sunshine. ISRO has recorded surprising temperature highs of 70° Celsius, much more than the 30°-40° it was expecting, and lows going down to -10°. We await many more fresh insights and nuances on aspects that will prove crucial for the future of space exploration.

The other uniqueness of this mission is that it was done on a comparatively meagre budget, less than the cost of a Hollywood space fantasy movie. Those who know how government organisations work, strangled by red tape, would appreciate ISRO's management systems that have packed in so much ingenuity, precision and cutting-edge knowhow. Also, the resilience to come back so triumphantly after bruising despair just four years ago. It is a tribute to the world-beating calibre of our scientists. Especially the mission approach the great scientist Satish Dhawan inculcated in the 1970s, upon taking over as ISRO chairman after Vikram Sarabhai died in 1971. This approach is what took Sarabhai's vision for Indian space and had it expanded and fulfilled—by setting defined objectives, building new centres to execute them, introducing timelines for delivery and instituting a rigorous, transparent review system that brought in even institutions outside ISRO. It is by thus cutting out the musty silo-based protocol that binds other sarkari departments that ISRO has now tasted success on the very frontiers of space exploration. That too at an extremely significant cusp period in the story of the global space race, when the moon is being seen not just as a destination but as a departure lounge for more distant forays.

In a more involved fashion than my school project, Group Editorial Director Raj Chengappa returns to his first love too. His first space cover story for INDIA TODAY came in 1984, about the stupendous feat achieved by the first Indian astronaut, Rakesh Sharma. Since then, he has written seven cover stories on space for us. This time, he travelled to Bengaluru, and had the privilege of visiting ISTRAC, the mission operation complex from where the scientists guided Chandrayaan-3 to its touchdown, and to the UR Rao Satellite Centre where Chandrayaan-3 was assembled. More importantly, he met key people who helmed the moon project, including its 46-year-old project director P. Veeramuthuvel and his 49-year-old deputy Kalpana Kalahasti. The most striking thing about them? They and their team are mostly small-town folks. Muthuvel is from Villupuram, Tamil Nadu, and did his M.Tech. from Trichy before a mid-career Ph.D. from ISRO took him to IIT Madras. Kalpana, whose ancestral roots are in Andhra Pradesh, studied engineering in a Chennai college before joining ISRO as a radar engineer at its launchpad in Sriharikota on the East Coast. "One thing that comes through with all of them is the sheer passion and dedication," says Chengappa. "They are all simple people handling the most complex science. There is no fluff in them, no tone of exaggeration or grandstanding, no sense of hierarchy either." What distinguished their approach is exemplary teamwork and coordination. About 20-30 departments were hands-on in ensuring utmost quality control in the three key systems—propulsion, navigation and guidance control. Each mind was also involved in taking the previous failure and stripping it bare in minute analysis. Each voice was heard. Thus, they turned failure itself into the foundation of success. Building in high degrees of redundancy, allowing more space for any tiny aspect that could go wrong.

I confess when the first moon mission was declared 20 years ago, I was cynical about its utility for a poor country so starved of resources and dogged by so many primary issues like child malnutrition and woeful lack of basic amenities. India is in a much better place today, and I have now changed my mind. Considering the scale of losses in government-run enterprises that the taxpayer has to bear otherwise and looking at the future, ISRO is definitely a worthwhile endeavour for us. It is a cost worth bearing. Also, it's a source of inspiration for scientists across the country and fosters high-tech industry, especially since space has now been declared open to private enterprise. More strength to their elbow for their future missions and I salute the multi-disciplinary talent of our scientists who make our space exploration possible. They do us proud.



(Aroon Purie)